

3ST Project Review

Consolidated gap analysis | 2026-07-09

Stack: Python FastAPI backend + Pixel Perfect UI (React/Vite) + Kite Connect. Strategy: 3ST (Triple SuperTrend + ADX). This document consolidates review findings for configure / backtest / live workflow completeness.

1. What Is Working

- Stock Selection with strategy settings (ST method, ST1/ST2/ST3 toggles, ADX, SL/TGT/TSL, session mode)
- Settings persist via selection store and flow into backtest when use_selection=true
- Kite authentication and instrument search
- Backtest: Kite up to ~400 days intraday, date pickers, points PnL, long/short split, start open / end close
- Watchlist queue lifecycle: waiting -> triggered -> active -> closed
- Signal scan: 3ST entry detection on waiting watchlist items
- Options spread preview and save (legs stored; preview API working)

2. Critical Gaps (Blocks Real Trading)

2.1 No Live Order Execution

Broker adapters exist (broker/kite_broker.py, broker/paper_broker.py) but nothing calls place_order(). POST /watchlist/{id}/activate only updates status. Live Desk Activate does not place orders even when ARMED.

Flow today: Scan -> triggered -> [manual Activate] -> active -> NO orders.

2.2 Step 7 - Options Spread Execution Missing

Spread legs are previewed and stored, but multi-leg order placement is not implemented. Backtest always runs on underlying OHLC, not spread PnL. watchlist_runner resolves index token for signals only; spread templates/legs ignored after trigger.

2.3 Live vs Backtest Logic Mismatch

Feature	Backtest	Live scan	Streamlit
Session / force exit	Yes	No	Partial
SL / TGT / TSL	Yes	No	Partial
Zone exits (ST1)	Yes	No	Partial
trade_mode Long/Short/Both	Yes	No	Yes
system_mode Intraday/Positional	Yes	No	Partial
ST enable flags	Yes	Entries only	No
Options spread product	No	Index signal only	No

Live scan uses strategies/three_st.py entry on last bar only. Backtest uses full backtest_engine with session, risk exits, and trade mode. Results can disagree materially.

2.4 Risk Limits API / UI Broken

UI field (settings.tsx)	API field (risk/limits.py)
max_loss_day	max_daily_loss

max_trades_day	(not defined)
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UI POSTs wrong field names; limits may not save. check_order() in risk/limits.py is never invoked.

3. Important Gaps (Polish & Consistency)

Hybrid ST method not implemented

UI offers 'Hybrid (HA ST, regular close)' but strategy_3st.py treats hybrid same as heikin_ashi.

Streamlit app.py is stale

Legacy Yahoo-only UI missing ST toggles, Kite source, new metrics, ATR TSL, date/Kite limits.

Session start/end not editable in UI

session_start/session_end stored but only force_exit shown on Stock Selection page.

Scan errors hidden

/watchlist/scan returns errors[] per item; Dashboard never displays them.

Options spread validation weak

Can save queue items with empty legs/expiry; no server check that preview was run.

Auth UX gap

API has /auth/login redirect; UI only supports manual request_token paste.

Port / env confusion

UI api.ts defaults port 8000; .env uses 8001; easy misconfiguration.

No automated tests

No project tests/ directory; strategy, API, spread builder untested.

Watchlist auto-activate not implemented

Scan promotes waiting->triggered only; manual Activate required on Live Desk.

Paper broker LTP

PaperBroker needs manual set_ltp(); positions stay empty without it.

4. Nice-to-Have

- Backtest price/ST chart (API returns candles; UI shows equity/trades only)
- Server-side scan scheduler (browser polling only when tab open)
- trade_mode not persisted in selection (resets on Backtest page)
- BANKNIFTY spread: index_token_key is None in config.py
- iron_condor in UI with no 4-leg execution path
- Root README with run commands and port alignment
- CORS allows * with credentials (production concern)
- Duplicate watchlist entries not prevented
- Firststock client in requirements appears unused by main API

5. Recommended Priority

Priority	Task	Effort
P0	Fix max_daily_loss / max_trades_day field mapping	Small
P0	Align live scan with backtest engine (session, exits, SL/TGT/TSL)	Medium
P1	Wire Activate + ARMED -> place_order() for underlying	Medium
P1	Implement or remove hybrid ST method	Small
P1	Show scan errors on Dashboard	Small
P2	Multi-leg spread orders (Step 7)	Large
P2	Options-spread backtest path	Large
P3	Update or deprecate Streamlit app.py	Medium

P3	README + port standardization	Small
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6. Summary

The research / configure / backtest loop is solid. The execute loop is not: scan -> trigger -> activate does not place orders, risk settings may not apply, and live signal logic is simpler than backtest.

Highest-impact path to live trading: (1) fix risk limits naming, (2) make live scanning use same rules as backtest, (3) wire Activate + ARMED to place_order().

Generated from codebase review of api/main.py, backtest_engine.py, strategy_3st.py, execution/, Pixel Perfect UI/src/, app.py, watchlist_store.py, selection_store.py, risk/limits.py.